

Get it back

Name	Group	Date
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The challenge In your beaker: 8 grams of salt, 50 millilitres of water, and a spoonful of sand, all mixed together. Get the salt back — all of it, dry, in the dish. You will be weighed.

FILTERING	DECANTING
Pour the whole mixture through a paper cone in a funnel. The sand stays on the paper, the liquid runs through.	Let the sand settle to the bottom, then carefully pour the liquid off it into a second beaker. No paper at all.

Half the class uses each one. We are finding out WHICH METHOD LOSES LESS SALT.

My method: FILTERING / DECANTING (circle one)

1 • Before you plan

Which of the three things in the beaker is dissolved, and which is not?

Why does that difference matter for getting the salt out?

A funnel on its own, with no paper in it — would that separate anything? Why not?

Which method do you think will lose more salt — and WHERE would it be lost?

What is the ONE thing you have to do first — and why does the order matter?

2 • Our method

Number the steps. Someone from another group must be able to follow it without asking you anything.

Do not forget Somewhere in your method there must be a step that says WEIGH THE EMPTY BASIN. If you do not know what the dish weighs by itself, you cannot work out how much salt ends up in it.

3 • Another group read our method

The question they asked:

What we changed because of it:

4 • Running it

Look at what came through the filter. The liquid is clear — but is it just water?

What is caught on the filter paper?

Why did the sand get caught but the salt did not?

5 • The numbers

Cool it first A hot basin gives a false reading — the warm air rising off it pushes up against the pan. Five minutes on the mat before it goes near the scales.

Empty basin (weighed at the start)	
Basin with dry salt in it	
SALT RECOVERED (subtract)	
Salt we started with	8 g
DIFFERENCE (how much is missing)	
Our method	filter / decant

Look at what is in the dish. What is it — and how do you know?

Did the missing salt get destroyed? How do you know?

6 • Which method won?

Nobody got 8 grams back. Both methods lost salt — but in DIFFERENT PLACES.

The whole class

Copy every group's recovered mass from the board.

Method	Group A	Group B	Group C
FILTERED (g)			
DECANTED (g)			

Which method got more salt back?

Was that what you predicted? If not, why did you think the other one?

Name the exact place each method loses its salt

Filtering loses salt...	
Decanting loses salt...	

Two groups used the SAME method and got different numbers. Does that mean one of them did it wrong?

What ONE change would improve EITHER method?

The big idea Neither method is wrong. They lose salt in different places, and until you compared them you had no way of knowing which place costs more. A number being off does not mean somebody was careless — it means the METHOD lost some, and the job is to say where.

Push me Work out both methods' recovery as a percentage of 8 g. How big is the gap between them? Is that gap bigger than the difference between two groups using the SAME method — and what would that tell you?

7 • Plant resin — the part that matters

Across Australia, First Nations peoples made adhesives from plant resin, including resin from spinifex and from grass trees. Warm it and it softens. Let it cool and it sets hard. It holds a stone point onto a spear shaft, a blade into a handle, a barb onto a hook.

Years later, out on Country, a long way from anywhere, the tool breaks. You do not need a new one. You warm the resin again, it softens again, you reset the point, and it hardens again — the same material, as many times as you need.

1 • Is softening and resetting resin a reversible or an irreversible change? How do you know?

2 • Which change we have studied is it most like — melting chocolate, or toasting bread? Say why.

3 • Why would being reversible be exactly the property you would want in a glue?

4 • Superglue cannot be undone. When would that be better, and when would it be worse?

Put the method in order

Name	Group	Date
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Cut out the steps. Put them in the right order. WARNING: two of them do not belong at all — find them and leave them out.

Pour the filtered liquid into the weighed basin.
Let the basin cool for five minutes.
Fold the filter paper into a cone and sit it in the funnel. (filtering)
Stir the sand until it dissolves.
Subtract the empty basin mass to find the salt recovered.
Hand the basin to the teacher to go on the frypan.
Weigh the empty basin and write the number down.
Weigh the basin with the salt in it.
Pour the salt down the sink and start again.
Pour the mixture slowly through the filter, into the second beaker. (filtering)
Wait for the water to evaporate away.
Let the sand settle, then pour the liquid off it carefully. (decanting)

My results

Name	Group	Date
Empty basin		
Basin with salt in it		
Salt I got back		
Salt I started with	8 g	
How much is missing		

Finish these sentences

My method was **FILTERING** / **DECANTING** (circle one)

The sand was separated out because _____
_____.

The salt went straight through because _____
_____.

The white solid in my dish is _____.

I did not get all 8 grams back because _____
_____.

The missing salt was NOT destroyed. It is _____
_____.

Word bank dissolved • too small • filter paper • left in the beaker • evaporated • salt • still damp • source of error

Exit tickets — cut into six

Name: _____ Both groups started with 8 g of salt. The group that FILTERED got 7.5 g back. The group that DECANTED got 6.5 g. <i>Neither group did anything wrong.</i> Explain how that is possible. _____ _____ _____	Name: _____ Both groups started with 8 g of salt. The group that FILTERED got 7.5 g back. The group that DECANTED got 6.5 g. <i>Neither group did anything wrong.</i> Explain how that is possible. _____ _____ _____
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